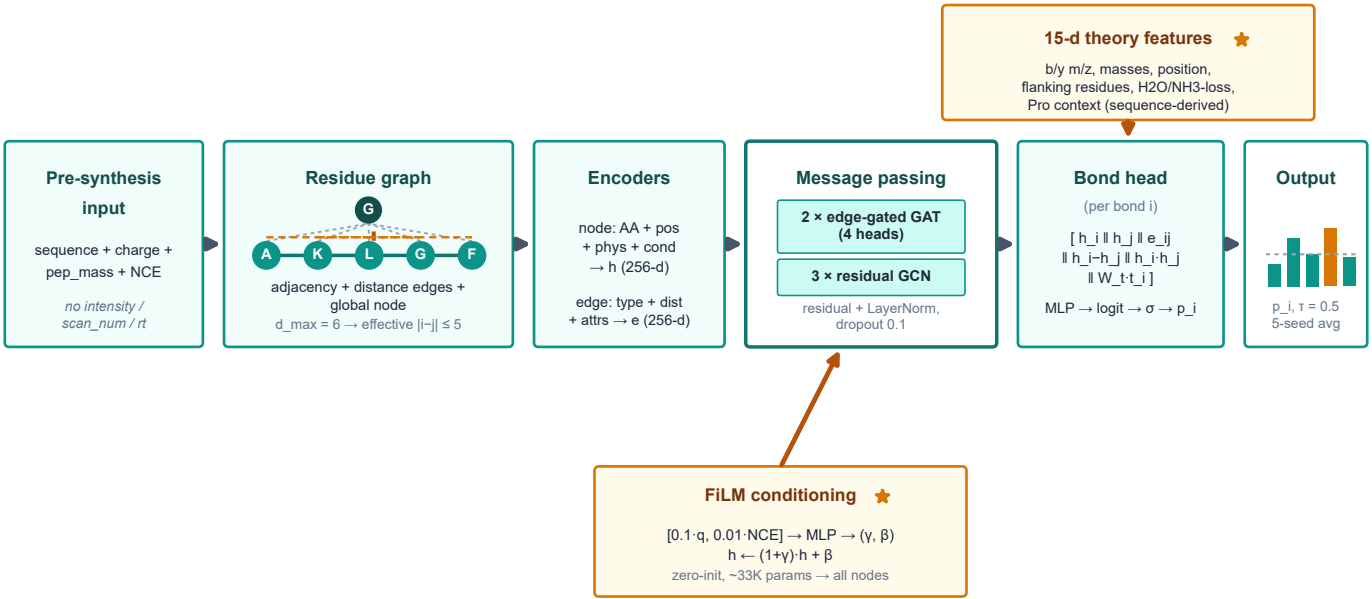

15-d theory features
b/y m/z, masses, position,
flanking residues, H2O/NH3-loss,
Pro context (sequence-derived)
Pre-synthesis input
sequence + charge +
pep_mass + NCE
no intensity /
scan_num / rt
Residue graph
G
A
K
L
G
F
adjacency + distance edges +
global node
d_max = 6 → effective |i−j| ≤ 5
Encoders
node: AA + pos
+ phys + cond
→ h (256-d)
edge: type + dist
+ attrs → e (256-d)
Message passing
2 × edge-gated GAT
(4 heads)
3 × residual GCN
residual + LayerNorm,
dropout 0.1
Bond head
(per bond i)
[h_i ‖ h_j ‖ e_ij
‖ h_i−h_j ‖ h_i·h_j
‖ W_t·t_i]
MLP → logit → σ → p_i
Output
p_i, τ = 0.5
5-seed avg
FiLM conditioning
[0.1·q, 0.01·NCE] → MLP → (γ, β)
h ← (1+γ)·h + β
zero-init, ~33K params → all nodes